

CE 310 — Week 9 Assignment: Simple Linear Regression

Week: 9 | **Topic:** Simple Linear Regression — Diagnostics and Application

Points: 60 pts total (3 problems)

Due: Wed Oct 28, 9:00 AM — submit to D2L

Overview

The in-class exercise built a simple linear regression model predicting hourly cooling energy from outdoor temperature for a DOE Reference Medium Office building in Tucson. This assignment goes beyond the exercise by examining the model's **limitations**, applying it to **track-specific design decisions**, and asking you to reflect critically on when a single linear model is — and is not — sufficient.

You will need: your completed exercise notebook (Wk09.03_In_Class_Exercise.ipynb), the DOE dataset (Week5_DOE_MedOffice_ZoneTemp_2023.csv), and a Python environment (Google Colab or local). **CE-track students need one more file** for Problem 2: CE310_BidProjects_2024_2026.csv, also on D2L.

How to Submit

Read this before you submit. Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
File format	Download as .ipynb — File → Download → Download .ipynb (not PDF, not .py)
Major declaration	Set MAJOR = "CE" or MAJOR = "ArcE" in the Setup cell — this determines which Problem 2 track is graded
Print lines	Do not edit or delete any <code>print(f"ANSWER_... = ...")</code> line
Written responses	Put each in its own markdown cell immediately after the code cell it responds to
Name and NetID	Fill in NAME and NETID in the identity cell and run it — that cell is what matches your work to your student record. A blank NETID takes a 5-point deduction.
File name	CE310_W9_Assignment_<NetID>.ipynb e.g. CE310_W9_Assignment_abc123.ipynb
Submission	Upload the .ipynb file to D2L → Assignments by 9:00 AM, Wed Oct 28

Setup cell (run first):

```
NAME = "" # e.g. "Jordan Reyes"
NETID = "" # e.g. "abc123"
print(f"NAME = {NAME}")
print(f"NETID = {NETID}")

import pandas as pd
```

```

import numpy as np
from scipy import stats

MAJOR = "CE"    # or "ArcE" – declares your Problem 2 track

df = pd.read_csv("Week5_DOE_MedOffice_ZoneTemp_2023.csv")
slope, intercept, r_value, p_value, std_err = stats.linregress(
    df['Outdoor_Temp_F'], df['Cooling_kWh']
)
print(f"MAJOR = {MAJOR}")

```

Problem 1 — Regression Diagnostics (30 pts)

Do not repeat the exercise computations. Use the slope, intercept, r , R^2 , and residual plot from your completed exercise notebook as starting points.

P1a (8 pts, written) — Residuals at Low Temperature

In your B8 residual plot, all residuals at outdoor temperatures below approximately 45°F are **positive** (the model predicts positive cooling energy, but the actual value is 0 kWh because the chiller is off).

In approximately 75 words, in a markdown cell, answer both of the following:

1. What does this systematic positive-residual pattern reveal about the validity of the linear model at low outdoor temperatures?
2. What type of regression model (not simple linear) might produce better predictions across the full temperature range? Name the approach and briefly explain why it would improve fit.

Graded on accuracy of model critique and appropriateness of proposed alternative.

P1b (8 pts) — Chiller Capacity Check

The building's chiller has a rated capacity of **200 tons** (1 ton of refrigeration = 3.517 kWh/hr of cooling).

Using your regression model (slope β_1 and intercept β_0 from the exercise):

(a) Predicted cooling demand at the mean outdoor temperature of 69°F

```

ANSWER_P1b_pred_69 = ____
print(f"ANSWER_P1b_pred_69 = {ANSWER_P1b_pred_69}")

```

(b) Remaining capacity above the 69°F baseline prediction

```

chiller_capacity = 200 * 3.517
ANSWER_P1b_remaining = ____
print(f"ANSWER_P1b_remaining = {ANSWER_P1b_remaining}")

```

(c) Additional degrees F needed to exhaust remaining capacity

```

ANSWER_P1b_deltaT = ____
print(f"ANSWER_P1b_deltaT = {ANSWER_P1b_deltaT}")

```

(d) Saturation temperature = 69 + deltaT

```

ANSWER_P1b_saturation_temp = ____
print(f"ANSWER_P1b_saturation_temp = {ANSWER_P1b_saturation_temp}")

```

In a markdown cell (2–3 sentences): does this saturation temperature exceed the maximum observed outdoor temperature in the dataset ($\approx 106^\circ\text{F}$)? What does this imply about whether the 200-ton chiller can ever be overwhelmed by Tucson's climate?

P1c (7 pts) — Summer-Only Regression

Run a separate simple linear regression using **only the summer rows** (Season == 'Summer').

```
summer_df = df[df['Season'] == 'Summer']
ANSWER_P1c_slope = ____
ANSWER_P1c_intercept = ____
ANSWER_P1c_r2 = ____
print(f"ANSWER_P1c_slope = {ANSWER_P1c_slope}")
print(f"ANSWER_P1c_intercept = {ANSWER_P1c_intercept}")
print(f"ANSWER_P1c_r2 = {ANSWER_P1c_r2}")
```

In a markdown cell: Is Summer R^2 higher or lower than the all-year R^2 ? Provide a physical explanation for the direction of change. (Hint: consider what other variables vary during summer but are relatively constant in a summer-only subset.)

P1d (7 pts) — Where the Line Stops Being Trustworthy

Your Week 15 written exam will ask you to judge a prediction made outside the data the model was fitted on. This is that question, on your own numbers.

(i) The fitted range and a prediction beyond it (2 pts).

```
ANSWER_P1d_fit_max = ____ # the highest Outdoor_Temp_F in the dataset, 1 decimal
ANSWER_P1d_pred_118 = ____ # your all-year model's prediction at 118°F, 2 decimals
print(f"ANSWER_P1d_fit_max = {ANSWER_P1d_fit_max}")
print(f"ANSWER_P1d_pred_118 = {ANSWER_P1d_pred_118}")
```

118°F is roughly Tucson's record high. The model will return a number for it without complaint.

(ii) Is the line even right at the edge? (2 pts).

Before asking whether to trust the model *past* its range, check how it does *at* the top of that range. Compute the mean residual — actual minus predicted — for every hour at or above 100°F:

```
ANSWER_P1d_resid_hot = ____ # mean residual for Outdoor_Temp_F >= 100, 2 decimals
print(f"ANSWER_P1d_resid_hot = {ANSWER_P1d_resid_hot}")
```

(iii) How much do the extreme hours pull the line? (1 pt).

Drop the hottest 1% of hours and refit the all-year model:

```
cut = df['Outdoor_Temp_F'].quantile(0.99)
ANSWER_P1d_slope_trimmed = ____ # slope after dropping hours above `cut`, 4 decimals
print(f"ANSWER_P1d_slope_trimmed = {ANSWER_P1d_slope_trimmed}")
```

(iv) Written (2 pts). In 4–6 sentences, put those three results together.

You should find something uncomfortable: the model **under-predicts** badly at the top of its range, yet dropping those extreme hours barely moves the slope. Reconcile that. Why can 80-odd hours be the ones the model fits worst and still have almost no power to change it? Then answer the design question: an engineer asks you for the cooling load at 118°F and you hand over your P1d(i) number. What do you have to say alongside it — and would you be more or less worried if the residual in (ii) had come out negative instead?

Look at your own residual plot. The exercise already had you build one. The pattern you are being asked to explain is visible in it — residuals high at the cold end, dipping through the middle, climbing again at the hot end. P1a asked you about the cold end. This is the end that sizes equipment.

Problem 2 — Track Application (18 pts)

Complete only the track matching your declared **MAJOR**. CE → Track A. ArcE → Track B.

Track A — CE: Does the Engineer's Estimate Predict the Winning Bid? (18 pts)

Second dataset: CE310_BidProjects_2024_2026.csv — download it from D2L. One row per awarded highway project from the Texas DOT bid tabulations, 4,575 projects. You will use this same file again in Weeks 11 and 12.

Before a project is advertised, the agency's engineer prepares an independent cost estimate. Bids come in later. A district that is programming a year of work needs to know how the two relate: if the estimates run systematically high or low, every budget built on them inherits that bias.

This is the same `stats.linregress` you used all week, pointed at a different pair of columns. Work in **millions of dollars** throughout — the starter notebook gives you the conversion.

(A2-a, 6 pts) Fit `WinningBid_USD` (response) against `EngineerEst_USD` (predictor), both in \$M, and report the line:

```
ANSWER_A2a_slope = ____
ANSWER_A2a_intercept = ____
print(f"ANSWER_A2a_slope = {ANSWER_A2a_slope}")
print(f"ANSWER_A2a_intercept = {ANSWER_A2a_intercept}")
```

(A2-b, 5 pts) Use the fitted line to predict the winning bid on a project the engineer estimated at **\$5.0M**, and report how much of the variation in winning bids the estimate explains:

```
ANSWER_A2b_pred_win = ____
ANSWER_A2b_r2 = ____
print(f"ANSWER_A2b_pred_win = {ANSWER_A2b_pred_win}")
print(f"ANSWER_A2b_r2 = {ANSWER_A2b_r2}")
```

(A2-c, 3 pts) Now answer the same question a second way, without the regression. Compute the bid ratio `WinningBid_USD / EngineerEst_USD` for every project, then report its mean and the percentage of projects awarded below their estimate:

```
ANSWER_A2c_mean_ratio = ____
ANSWER_A2c_pct_under = ____
print(f"ANSWER_A2c_mean_ratio = {ANSWER_A2c_mean_ratio}")
print(f"ANSWER_A2c_pct_under = {ANSWER_A2c_pct_under}")
```

Note: the intercept in A2-a comes out to essentially zero. That is worth noticing — it means the relationship is proportional, so the slope by itself is a bid ratio, directly comparable to A2-c.

(4 pts, written) In a markdown cell (3–4 sentences): the slope and the mean bid ratio are both “the average discount,” and they do **not** agree. Explain why the two differ — think about which projects carry the most weight in each calculation. Then say which of the two you would hand a district that is programming next year's construction budget, and why.

Track B — ArcE: ASHRAE 55 Adaptive Thermal Comfort (18 pts)

ASHRAE Standard 55 (*Thermal Environmental Conditions for Human Occupancy*) includes an **adaptive thermal comfort model** for naturally ventilated spaces. The comfort zone upper limit can be approximated as:

$$T_{\text{comfort, max}} = 0.31 \times T_{\text{outdoor, prevailing mean}} + 60.4 \text{ (}^{\circ}\text{F)}$$

(Equation: ASHRAE 55-2023 80% acceptability upper limit, $T(^{\circ}\text{C}) = 0.31 \cdot T_{\text{pma}}(^{\circ}\text{C}) + 21.3$, converted to $^{\circ}\text{F}$: $T(^{\circ}\text{F}) = 0.31 \cdot T_{\text{pma}}(^{\circ}\text{F}) + 32 \times 0.69 + 21.3 \times 1.8 = 0.31 \cdot T_{\text{pma}}(^{\circ}\text{F}) + 60.4$)

(B2-a, 6 pts, manual — plot required) Using this equation, compute $T_{\text{comfort_max}}$ for outdoor temperatures ranging from 40°F to 110°F in 1°F increments. Plot T_{outdoor} (x-axis) vs. $T_{\text{comfort_max}}$ (y-axis). Add a horizontal reference line at 80°F (a commonly cited upper limit for comfort compliance in mixed-mode buildings).

```
T_outdoor = np.arange(40, 111, 1)
T_comfort_max = 0.31 * T_outdoor + 60.4
# your plot code here
```

(B2-b, 6 pts) Identify the **outdoor temperature threshold** above which $T_{\text{comfort_max}}$ exceeds 80°F. Solve algebraically, then compute:

$$0.31 \times T_{\text{outdoor}} + 60.4 > 80 \implies T_{\text{outdoor}} > ?$$

```
ANSWER_B2b_threshold = ____
print(f"ANSWER_B2b_threshold = {ANSWER_B2b_threshold}")
```

Show the algebra in a markdown cell alongside the code.

(B2-c, 6 pts) Count how many hours in the Tucson dataset have an outdoor temperature above this threshold:

```
ANSWER_B2c_hours_above = ____
ANSWER_B2c_frac_above = ____
print(f"ANSWER_B2c_hours_above = {ANSWER_B2c_hours_above}")
print(f"ANSWER_B2c_frac_above = {ANSWER_B2c_frac_above}")
```

In a markdown cell: for each of those hours, the adaptive model predicts that an 80°F indoor limit would be exceeded in a naturally ventilated space. What does this imply for the feasibility of naturally ventilated offices in Tucson's climate?

Problem 3 — Written Reflection (12 pts, written)

Write a **100–150 word** reflection (no code required) in a markdown cell, addressing the following prompt:

Describe what your residual plot from B8 reveals about the validity of using a single linear regression model to predict cooling energy for this building throughout the entire year. Cite at least **two specific quantitative observations** from your analysis (e.g., a specific temperature range, a residual magnitude, an R^2 value, or a count of affected hours). Then propose **one modification** to the modeling approach that would address the largest structural limitation you identified.

Grading criteria (12 pts total): - Two quantitative observations cited (4 pts — 2 pts each) - Accurate interpretation of what the observations reveal about model validity (4 pts) - Proposed modification is specific, feasible, and addresses the identified limitation (4 pts)

Submission Checklist

- ☐ Setup cell run with MAJOR declared
- ☐ P1a: written response ~75 words
- ☐ P1b: all four ANSWER_ values printed + written discussion
- ☐ P1c: ANSWER_ slope/intercept/r2 printed + written comparison
- ☐ P1d: ANSWER_ slope/intercept/r2 printed + written comparison
- ☐ P2 (only your declared track): all ANSWER_ values printed, plot (Track B only), and written responses
- ☐ P3: reflection, 100–150 words, two quantitative observations, one proposed modification